

Will Cygan

SENIOR SOFTWARE ENGINEER, LINKEDIN BUSINESS PLATFORM

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Work Experience

Senior Software Engineer

Chicago, IL

LINKEDIN

Mar 2024 – Present

- Led design and implementation of real-time alerting system processing 100,000+ QPS using Kafka/Flink/Venice, recovering \$2M+ annually in involuntary churn by notifying subscribers of failed payments before subscription cancellations; platform extended by 3 partner teams (Ordering, Payments, Mobile) for Winback, Usage Billing, and Mobile Billing alerts.
- Automated Venice TTL purging of 35+ day records on the 100K QPS Global Alerts cache, cutting cache size 40% and downstream Invoice Search API QPS 50%.
- Designed Oracle-to-MySQL Delegation Framework adopted by all 10 LBP domain teams as the org-wide migration pattern, with novel Couchbase-TTL entity-routing cutover for strong consistency and GoldenGate-caught-up safety guarantees.
- Led full zero-downtime Oracle-to-MySQL migration of LinkedIn's Ordering Backend (12TB / 20B rows / 6 tables), rewriting 30,000+ lines of SQL across ~75 PRs with JooQ/Flyway; zero data loss through 6-week ramp validated at 10,000+ orders.
- Eliminated N+1 query problem in Order Processing system, reducing average query latency by 40% (50ms→30ms) and p95 by 37% (200ms→125ms) across all ordering workflow read paths.
- Built rate-limited distributed deletion pipeline (Airflow, Trino, Kafka, gRPC) that purged 300,000 corrupted order records in production; same pipeline now ramping toward an identified 33% reduction across billions of dormant legacy OMS rows in the 10+ year-old Ordering database.
- Designed Unified Optimistic Locking mechanism for the MySQL Ordering Backend, eliminating a class of silent conflicting-write failures under Temporal activity retries via version-based concurrency control across every update-based write path.
- Designed Context Repos pattern (git submodules + curated context files) for agentic code navigation across LBP's 30+ decomposed microservices; variant adopted by LinkedIn's Developer Productivity org in Feb 2026.

Software Engineer

San Francisco, CA

LINKEDIN

Feb 2022 – Mar 2024

- Scaled Videos You Might Be Interested In recommendation to 3,000 QPS on LinkedIn's 50M+ DAU Flagship Feed, driving +2.5% Weekly Skilled Learners (LinkedIn Learning's north-star metric) and +0.57% SWI on Skill Credits across 3 experiments.
- Built Learning Alerts V2 member/stage targeting framework from scratch, driving +1.09% WSL (Learning's north-star) via Spark pipelines over 20+ datasets (~50TB/run) classifying 10M+ job seekers into career-change-funnel cohorts with deterministic bucketing for clean A/B readouts; ramped safely to 22M members.
- Improved LinkedIn Flagship Feed engagement +0.2% by filtering audio-only content from video course recommendations, a statistically significant lift at hundreds of millions of daily impressions.

Projects

Anton (Kubernetes Homelab)

github.com/wcygan/anton

- Built fault-tolerant 3-node bare-metal Kubernetes cluster using Talos Linux with GitOps CI/CD automation, achieving 99.9% uptime over 12+ months.
- Deployed distributed data systems (TiDB, RedPanda, DragonflyDB, ScyllaDB, ClickHouse) to evaluate performance characteristics and operational trade-offs of modern database alternatives.

Skills

Languages Java, Rust, Python, SQL, Bash, TypeScript

Technologies Kafka, gRPC, Temporal, Flink, Spark, Airflow, Trino, Kubernetes, Docker

Databases MySQL, Redis, Couchbase, Oracle, DragonflyDB, Venice, HDFS, S3

Education

University of Illinois at Chicago

Chicago, IL

B.S. IN COMPUTER SCIENCE

2021